ConnectsApp: TikTok-Style Livestreaming Architecture Specification

Document Version: 4.0.0 (Clean Architecture Specification)
System Target: ConnectsApp Livestreaming Architecture
Core Model: Two-Tier Hybrid Architecture (WebRTC Interactive Stage + Metered HLS/LL-HLS Mass Delivery)
Core Technologies: Vanguard Media Engine + LiveKit WebRTC SFU + Low-Latency HLS (CMAF) + Soketi (WebSockets) + TigerBeetle (Financial Ledger) + Redis (Realtime Game State) + PostgreSQL/MySQL + Flutter
Target Latency: Interactive Stage <80ms; Mass Audience 1–3s; Chat/Reactions/Gifting <50ms

YELLOW HIGHLIGHTS **Team Notice:** All newly added product points (Section 15: Points 15.1 – 15.7) and ratified ledger/capacity modifications (§6.1, §6.2, §12.1) are highlighted in yellow for immediate team review.

1. Executive Summary & Design Principles

This document defines the dedicated production architecture for **ConnectsApp Livestreaming**. It specifies the media delivery, gamified battle mechanics, interactive co-hosting, and financial ledger pipelines required to achieve TikTok-style feature parity while controlling operational delivery costs and respecting mobile silicon limits.

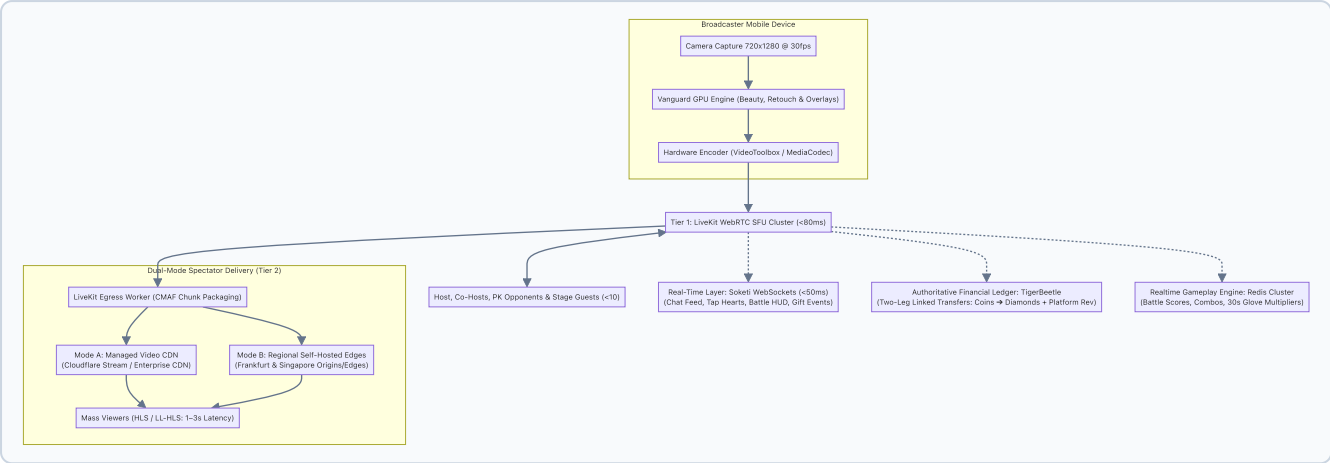
Core Architecture Principles

1. **Strictly Two-Tier Delivery:** WebRTC is strictly reserved for the broadcaster, co-host, PK battle opponent, and host-approved multi-guests (<10 active publishers). Mass spectators never connect to WebRTC; they consume cost-effective HLS/LL-HLS streams.
2. **Dual-Mode Delivery Strategy:** Video distribution supports two interchangeable modes:
    - **Mode A (Managed Video CDN):** Cloudflare Stream or Enterprise CDN for operational simplicity at launch.
    - **Mode B (Regional Self-Hosted Edges):** Cost-control path deploying NGINX edge caches on dedicated hardware in Frankfurt and Singapore first, expanding to Australia and North America later.
    - **Static Assets & VOD:** Cloudflare/R2 or equivalent object storage handles static assets, VOD recordings, clips, thumbnails, gift animations, and PAG files.
3. **Fixed Resolution & Frame Rate Baseline:** Broadcaster cameras, GPU processing, and hardware encoders are locked to **true 720×1280 portrait at 30 fps**. There is no 60 fps capture anywhere in the live pipeline.
4. **Authoritative Double-Entry Financial Integrity:** TigerBeetle is the sole authority for financial balances (Coins, Diamonds, Revenue). Redis stores only ephemeral, non-financial game state (battle score bars, combo streaks, timers).
5. **Prestige vs. Admission Separation:** Top contributors and VIPs receive visual prestige badges and queue priority only. Stage camera/mic admission always requires a 4-gate verification process.

2. End-to-End Livestreaming System Architecture

The following diagram illustrates the complete end-to-end dataflow from broadcaster capture to mass viewer playback:

FIGURE 1: END-TO-END SYSTEM ARCHITECTURE (TWO-TIER HYBRID DELIVERY)



3. Two-Tier Delivery, Dual-Mode CDN Strategy & ABR Ladder

3.1 Two-Tier Audience Model

- **Tier 1: Interactive Stage (WebRTC / LiveKit):**
  - **Participants:** Broadcaster, co-hosts, PK opponents, and approved stage guests (typically 1 to 4, hard cap of 10).
  - **Protocol:** WebRTC over UDP (ICE / DTLS / SRTP).
  - **Latency:** Glass-to-glass latency under 80 milliseconds.
  - **Bandwidth Usage:** ~1.4 Mbps upstream per publisher.
- **Tier 2: Mass Spectators (HLS / LL-HLS):**
  - **Participants:** 100 to 500,000+ viewers watching the broadcast.
  - **Protocol:** Low-Latency HLS with 500ms CMAF chunked transfer over standard HTTP/2 and HTTP/3.
  - **Latency:** 1.2 to 2.5 seconds perceived video latency.
  - **Real-Time Illusion:** Chat, floating hearts, and gift banners arrive via Soketi WebSocket (<50ms), creating an instantaneous interactive feeling even while video has a 1.5s buffer.

3.2 Delivery Cost Strategy: Dual-Mode Architecture

Video distribution is budgeted as a variable cost of goods sold (COGS). The architecture avoids vendor lock-in by supporting two interchangeable delivery modes:

Dimension	Mode A: Managed Video CDN	Mode B: Self-Hosted Regional Edges
Primary Use Case	Product launch, initial traction, operational simplicity.	Scale-out cost reduction when monthly viewer hours surge.
Technology	Cloudflare Stream, AWS IVS, or Fastly/CloudFront Enterprise CDN.	NGINX / OpenResty cache nodes on fixed-bandwidth dedicated servers.
Initial Deployment	Cloudflare Stream global edge network.	Frankfurt (Europe origin/edge) + Singapore (APAC edge).
Planned Expansion	N/A (Global coverage built-in).	Sydney (Australia) + Ashburn/Dallas (North America).
Billing Model	Metered per delivered minute (~$1.00/1,000 min) or per TB egress.	Fixed monthly dedicated server lease with unmetered 1 Gbps/10 Gbps uplinks.
Switching Mechanism	Dynamic playlist URL routing in backend stream coordinator.	Seamless fallback to Mode A if self-hosted capacity hits 85%.

3.3 Role of Static CDN & Object Storage

Regardless of the live video delivery mode, standard CDN and object storage (Cloudflare CDN, Cloudflare R2, or Hetzner Object Storage) are used for: * **VOD Archives:** Permanent stream recordings saved by LiveKit Egress. * **Instant Clips:** 30-second highlight clips generated for social sharing. * **Stream Thumbnails:** Automated snapshot previews captured every 10 seconds. * **Gift & Animation Assets:** SVG icons, Lottie files, sound effects, and libpag animation packages. * **Static App Assets:** Creator avatars, banner images, and static UI bundles.

3.4 Adaptive Bitrate (ABR) Ladder

To protect spectator playback on cellular networks and control outbound bandwidth, LiveKit Egress encodes four standard profiles:

Profile	Resolution	Frame Rate	Video Bitrate	Audio Bitrate	Target Audience / Condition
720p High	720 × 1280	30 fps	2,000 kbps	128 kbps	Wi-Fi, high-speed 5G, full-screen playback
576p / 540p Med	540 × 960	30 fps	1,100 kbps	96 kbps	Standard 4G/LTE mobile networks
360p Low	360 × 640	30 fps	550 kbps	64 kbps	Congested cellular networks, data-saver mode
180p Fallback	180 × 320	15 fps	180 kbps	32 kbps	Extreme edge connectivity, audio-priority fallback

4. VIP Ranking, Prestige & Stage Admission Boundaries

4.1 VIP Prestige Hierarchy

ConnectsApp rewards generous contributors with visual status and priority, but separates prestige from camera/microphone stage access:

- **Level 40+ Emperor:** Custom neon screen entrance banner, gold username in chat, VIP chat flair, priority queue position when requesting stage access.
- **Top 3 Room Contributors:** Avatars pinned on the stream HUD in gold, silver, and bronze slots; special badge in chat feed.
- **Monthly VIP Subscribers:** Dedicated badge, subscriber-only chat modes, exclusive emotes.
- **Regular Viewers:** Standard chat, tapping hearts, gifting from standard catalog.

4.2 Gated Stage Admission Engine (4-Gate Verification)

High spending never grants automatic camera or microphone broadcast rights. Admitting a user to the WebRTC stage requires passing all four gates:

1. **Gate 1: Viewer Request:** The viewer explicitly taps “Request to Join Stage”.
2. **Gate 2: Safety & Eligibility Check:** Backend automated check validates user age (18+), account standing (no active strikes), and device readiness.
3. **Gate 3: Broadcaster Approval:** The host receives an in-app prompt and manually accepts the request.
4. **Gate 4: Viewer Mutual Consent:** The viewer confirms camera/mic permissions before publishing their track to LiveKit.

4.3 Seamless Stage Promotion (Hot-Swap)

When an approved spectator transitions from spectator to stage guest: 1. Flutter client receives `room:promoted_to_stage` over Soketi WebSocket. 2. App requests a scoped LiveKit participant token with `canPublish: true`. 3. WebRTC client connects and acquires local camera/microphone. 4. Audio cross-fades over 150ms from the HLS stream to WebRTC while the HLS player unmounts. 5. The transition completes without reloading the stream room or losing chat history.

5. Video Pipeline, Thermal Management & Bitrate Optimization

5.1 Native Camera & Pipeline Invariant: 720p @ 30 fps End-to-End

- **Hardware Baseline:** Mobile camera sensors in ConnectsApp are physically negotiated and locked to **true 720×1280 portrait at 30 fps** (Slice E4 contract invariant).
- **Strict 1:1 Cadence Across the Entire Stack:** The entire pipeline operates synchronously at 30 fps from camera glass to subscriber screen:
    1. **Camera Sensor Capture:** 720×1280 @ 30 fps (rotation=0).
    2. **Vanguard GPU Engine:** Renders beauty retouching, green screen, and overlays at exactly 30 fps on native GPU textures.
    3. **Hardware Encoder:** VideoToolbox (iOS) or MediaCodec (Android) encodes frames directly from the GPU surface.
    4. **LiveKit SFU:** Relays 30 fps RTP packet streams without server-side transcoding.
    5. **Subscriber Viewports:** Decodes and renders at native 30 fps.

5.2 Thermal State Machine & App-Level Thermal Guard

- **Existing App Service:** Device temperature and thermal throttling are monitored centrally by `ThermalGuardService` (`lib/app/shared/media/thermal_guard/thermal_guard_service.dart`).
- **Livestream Dynamic Adaptation (`PLANNED` / `NOT_PROVEN`):** Automatically scaling down livestream video bitrate, reducing frame rates, or disabling full-screen gift overlays based on thermal events is designed as follows:
    - **Tier 0 (Normal):** Full 720p @ 30 fps (1.4 Mbps), full-screen gift animations enabled.
    - **Tier 1 (Moderate - Temp > 39°C):** Maintain 720p @ 30 fps (1.1 Mbps), throttle particle effects by 50%, limit concurrent animations to 1.
    - **Tier 2 (Severe - Temp > 43°C):** Downscale to 540p @ 30 fps (600 kbps), disable full-screen gift animations, display static gift badges only.
    - **Tier 3 (Critical - Temp > 46°C):** Drop to 180p @ 15 fps or audio-only, pause camera hardware, display static standby card to cool silicon.

5.3 Client-Side Whale Gift Animations: PAG / libpag Guardrails

- **Rendering Engine:** Tencent libpag (Portable Animated Graphics) provides high-performance vector/B-frame animations exported from Adobe After Effects.
- **Client-Side Overlay Only:** Gift animations are rendered locally on the client viewport as transparent overlays; they are never composited into the broadcaster's video encoder stream.
- **Mandatory Legal & Build Audit Gate:** Because libpag may bundle third-party LGPL dependencies, integration is subject to formal legal license audit and isolation before production build inclusion.
- **Performance Guardrails:**
    - Strict memory budget: Max 40 MB texture memory for cached animations.
    - Queue concurrency: Max 1 full-screen animation active at any time; lower-tier gifts fall back to 2D sprite/CSS animations.
    - Remote Kill-Switch: Feature flag allows instant server-side disabling of PAG rendering in case of memory leaks or crashes on specific Android GPU chipsets.

6. Financial Ledger & Realtime Engagement Engine

6.1 System Responsibility Map: TigerBeetle vs. Redis vs. SQL

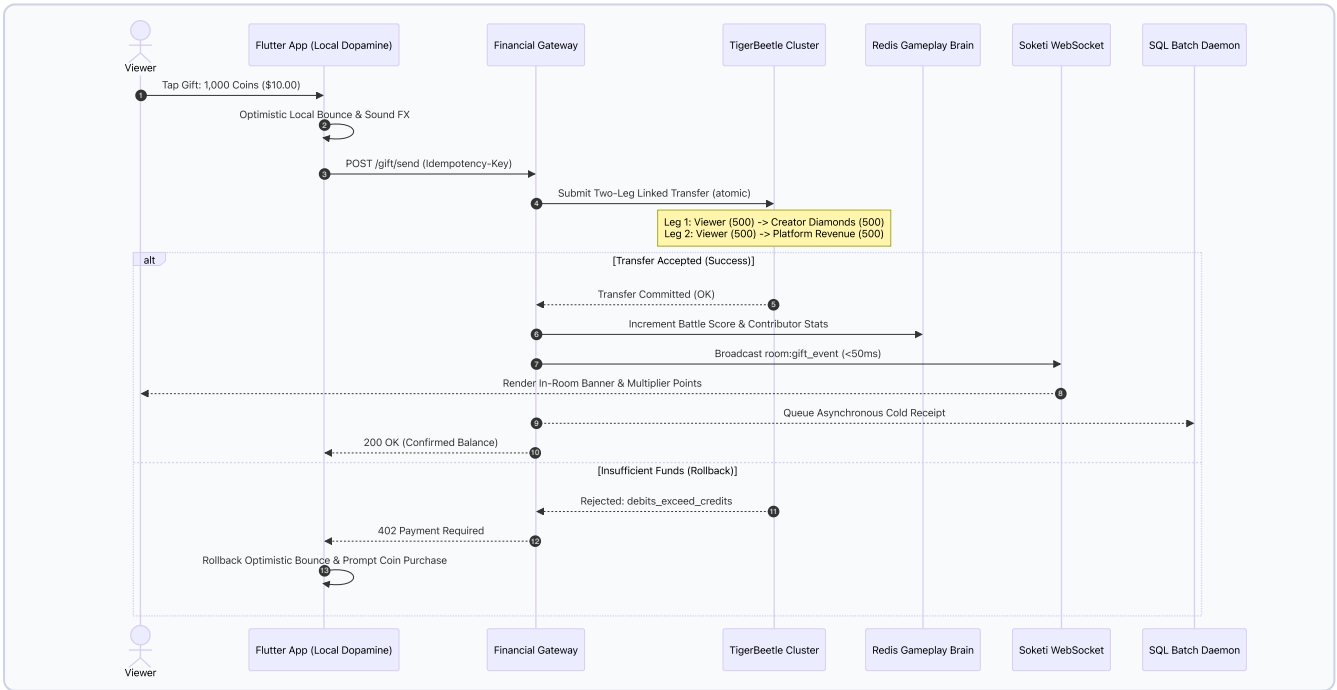
Responsibility Dimension	TigerBeetle (Authoritative Financial Core)	Redis Cluster (Realtime Ephemeral Gameplay)	PostgreSQL / MySQL (Application Data Store)
Primary Role	Financial balances, double-entry ledgers, transaction immutability.	Ephemeral game state, battle scores, combo streaks, timers.	User profiles, gift catalogs, stream metadata, historical reporting.
Data Handled	User Coin balances, Creator Diamond liabilities, Platform revenue.	PK match scores, MVP rankings, 30s glove multipliers, tap counters.	Purchase receipts, transaction audit logs, moderation records.
Overdraft Protection	Enforced via `flags.debits_must_not_exceed_credits` (100% ACID).	None (Ephemeral in-memory cache; resets after match ends).	Standard relational foreign keys and transaction boundaries.
Throughput Target	UPDATED Designed for high-throughput financial transfer workloads; ConnectsApp benchmark target remains NOT_PROVEN until tested in our environment.	Sub-millisecond latency for WebSocket broadcast pushes.	Standard batched writes and read replicas.

6.2 Authoritative TigerBeetle Ledger Flow

Every virtual gift transaction follows strict double-entry accounting with explicit linked transfers:

RATIFIED TRANSFER MODEL **Balanced Double-Entry Accounting:** Leg 1: Debit Viewer Coin Account (500) ➔ Credit Creator Diamond Account (500) | Leg 2: Debit Viewer Coin Account (500) ➔ Credit Platform Revenue Account (500). Total viewer debit = 1,000 coins ($10.00). Total credits = 500 diamonds ($5.00) + 500 platform revenue ($5.00). Both legs are linked and execute atomically.

FIGURE 2: AUTHORITATIVE TIGERBEETLE DOUBLE-ENTRY LEDGER FLOW



6.3 Two-Ledger Architecture: Battle Points vs. Cash Diamonds

To prevent financial exploits while enabling high-energy game mechanics, ConnectsApp completely decouples gameplay points from real money:

- **The Real-Money Ledger (TigerBeetle):**
  - When a viewer spends 1,000 Coins ($10), the creator receives exactly **500 Diamonds ($5)**.
  - Diamonds represent legal financial liabilities. **Diamonds are never multiplied by gameplay power-ups**.
- **The Game Score Bar (Redis):**
  - Gifts increment the in-memory battle score bar.
  - During bonus events (Speed Challenge 2x, Booster Glove 5x), points on the score bar are multiplied (e.g., 1,000 Coins ➔ 5,000 Battle Points).
  - **Battle points have $0 cash value** and reset to 0 the moment the battle concludes.

---

7. Real-Time Chat, Multi-Guest & PK Battle Protocols

7.1 Real-Time Chat Architecture

- **Engine:** Powered by Soketi (Pusher-compatible, open-source WebSocket server written in C++ via µWebSockets).
- **Delivery:** Delivers chat messages, room events, and reaction counts to viewers in under 50 milliseconds.
- **Rate Limiting:** Sliding-window rate limiter restricts users to 3 messages per 5 seconds.
- **Dynamic Throttling Under Surge:** If a room exceeds 10,000 viewers, client chat rendering transitions to message-sampling (rendering top VIP messages and a smoothed sample of general chat) to prevent mobile UI thread jank.

7.2 TikTok LIVE Match & Co-Host Architecture

7.2.1 Two Initiation Paths

1. **Path 1: Manual Match (Friend / Follower Challenge):**
   - Two creators enter a 50/50 split-screen co-host session via LiveKit cross-subscription.
   - Either creator taps the Match (🥊) icon to challenge the other.
   - Once accepted, a synchronized 3-second countdown starts.
2. **Path 2: Quick Match (Algorithmic Matchmaking):**
   - A creator broadcasting solo selects Quick Match.
   - Backend pairs them with another creator searching in the matchmaking pool.
   - Streams instantly cross-subscribe and launch into the 3-second countdown.

7.2.2 The 5-Minute Battle Lifecycle

The battle lifecycle strictly mirrors TikTok LIVE battle mechanics:

- **Phase 1: Sync Countdown (3s):** Server broadcasts `room:pk_countdown` with NTP timestamp. Both screens play a 3-2-1 opening battle animation.
- **Phase 2: Battle Period (300s / 5 Minutes):**
  - Split-screen view stays locked; dynamic tug-of-war score bar mounts above the video.
  - Room A gifts increment Host A points; Room B gifts increment Host B points.
  - Soketi pushes score bar updates every 500ms with top contributor avatars.
  - Opening 30–60s features the Speed Challenge bonus target.
- **Phase 3: Winner Announcement (5s–10s):**
  - Scores freeze; Victory 🏆 and Defeat overlays are displayed.
  - MVP and top 3 contributors for each team are highlighted.
  - Tug-of-war bar unmounts; split-screen co-host layout remains seamlessly active.
- **Phase 4: Post-Match Co-Host & Creator-Directed Punishment (Open-Ended / Zero Forced Cutoff):**
  - **NO SYSTEM TIMER:** The system does NOT enforce a 60s or 120s timer on forfeits.
  - **Creator-Governed Duration:** The creators decide what the forfeit is and how long it lasts. The loser performs the forfeit live on camera while the winner celebrates and banters.
  - **Post-Battle Options:** The creators remain together in split-screen until they choose an action:
    - **Rematch:** Either creator taps “Rematch” ➔ once accepted, starts the next 5-minute battle.

- **Social Hangout:** Creators stay in split-screen to chat, collaborate, and broadcast together.
- **Clean Disconnect:** Either creator taps “Disconnect” (X) at any time. The split-screen smoothly closes, and both solo livestreams continue uninterrupted with their own audiences.

7.2.3 2v2 Squad Battles (Team Matches)

ConnectsApp supports 2v2 Team Battles connecting 4 creators in a 2×2 split-screen grid: * **Independent Ingress:** All 4 creators broadcast from their own rooms to their respective audiences. * **Cross-Subscription:** Each room subscribes to the other 3 video tracks via LiveKit SFU. * **Team Point Pooling:** Redis aggregates points under `match:{id}:team_blue` and `match:{id}:team_red`. * **Individual Diamond Accrual:** Although points pool for the team score, **100% of coins and diamonds accrue to the individual creator who received the gift**.

7.2.4 Selective Opponent Audio Mute (One-Way Local Mix Mute)

Each host has an on-screen “Mute Opponent” toggle: * **Host A Taps Mute:** LiveKit drops Host B’s audio track *only* for Room A subscribers and Host A’s handset. * **Result:** Host A is 100% audible to Audience A, but Audience A hears zero audio from Host B. This enables tactical pep-talks (*“Save your gifts for the last 15 seconds!”*) or playing custom soundboard music. * **Asymmetric Muting:** Host B is not muted; Audience B continues hearing both creators. * **Instant Unmute:** Tapping Unmute re-subscribes Host B’s audio into Room A instantly without crackle.

7.2.5 Hardware-Level Zero-Echo Audio (WebRTC AEC3)

Even with creators speaking on open phone speakers without headphones: * WebRTC engages native hardware Acoustic Echo Cancellation (`VoiceProcessingIO` on iOS, `AcousticEchoCanceler` on Android). * The speaker output buffer is fed directly back into WebRTC’s AEC3 engine as a reference signal, phase-inverted, and subtracted in under 10ms.

7.2.6 Tapping Economy & Tactical Power-Up Arsenal

- **Screen Tapping (3 Points/Tap):** Non-paying viewers double-tap the screen to award 3 battle points per tap to their creator (capped at 300 points per match per user to prevent autoclickers).
- **Speed Challenge (First 30–60s):** Fast tapping and small gifts (Roses) hit a speed milestone to unlock **Doubles (x2 points)** for 30 seconds.
- **Booster Glove:** Earned exclusively by winning MVPs and stored in their in-app Backpack (5–7 day expiration). When deployed, grants a **30% chance for a 5x Critical Hit** on battle points for 30 seconds.
- **Magic Mist (Fog of War):** Blurs and conceals the team’s score from opponents for 30 seconds, enabling dramatic last-second snipes.
- 🔥 **Win Streaks:** Consecutive match wins display a flaming badge. Sniping high win streaks drives intense community defense spending.
- **Treasure Boxes:** Whales or creators drop coin chests with 2–3 minute countdown timers. Viewers stay glued to the stream to claim random coin shares, breaking friction for first-time gifting.

---

8. Safety, Compliance, & Moderation Architecture

- **Age Verification & Minor Protection (COPPA / GDPR-K):** Broadcasters must be verified 18+. Spectators under 18 have gifting features locked.
- **Guest Audio/Video Moderation:** Hosts retain sovereign control with one-tap ability to mute, hide, or kick any stage guest or PK opponent.
- **Audit Trail:** Moderation events (kicks, mutes, warnings) are logged with 60-second video snippet buffers stored in object storage for compliance review.
- **Gifting Velocity Limits (AML):** Unverified accounts capped at $250/day. Rapid transactions trigger automated risk scoring and cooling-off locks.
- **Regional Policy Geofencing:** In jurisdictions where RNG power-ups fall under gambling regulations, the Booster Glove’s 30% chance is replaced with a deterministic, fixed point boost.
- **Automated Moderation Pipeline (`PLANNED` / `NOT_PROVEN`):** Egress optical frame extraction sampling 1 frame every 3 seconds for AI nudity, violence, and OCR text scanning.

---

9. LiveKit Egress Packaging & Stream Recording (VOD)

Under this architecture, LiveKit Egress performs dual duties: 1. **Low-Latency HLS Packaging:** Packages 720p 30fps video into 500ms low-latency CMAF chunks (`master.m3u8`) uploaded directly to Mode A CDN or Mode B regional edges. 2. **Permanent VOD Storage:** Simultaneously records the raw MP4 stream to S3-compatible object storage (e.g., Hetzner Object Storage at €0.01/GB/month) for permanent replay and creator clips.

10. Security & Token Authentication Architecture

- **Scoped LiveKit JWT Tokens:** Tokens are signed server-side using HS256 with strict permission flags:
  - **Broadcaster Token:** `{ "room": "room_id", "canPublish": true, "canSubscribe": true, "canPublishData": true, "exp": 14400 }` (4 hours).
  - **Stage Guest Token:** `{ "room": "room_id", "canPublish": true, "canSubscribe": true, "canPublishData": false, "exp": 1800 }` (30 minutes).
  - **Spectators:** Connect only to HLS endpoints; they receive no LiveKit WebRTC tokens.
- **Rate Limiting:** Maximum 5 room token generations per user per minute enforced via Redis Token Bucket.

11. Monitoring, Alerting & Observability

- **Telemetry Collection:** Prometheus scrapes LiveKit SFU metrics (`/metrics`), Node Exporter (CPU, RAM, NIC I/O), Redis Exporter, and Soketi WebSocket metrics.
- **Central Dashboards:** Grafana provides real-time visibility into active rooms, stage participant counts, egress encoding latencies, and CDN origin bandwidth.
- **Alerting Rules:**
  - **P0 Alert:** Node packet loss > 2.5% for 60 seconds ➔ Immediate dispatch.
  - **P0 Alert:** LiveKit SFU unreachable ➔ Cloudflare DNS auto-failover + on-call alert.
  - **P1 Alert:** Redis memory utilization > 80% ➔ Automatic flush of expired keys.
  - **P1 Alert:** Egress encoding backlog > 3 jobs ➔ Auto-scale additional egress worker.

12. Scalability & Reality-Grounded Cost Model

12.1 Server Fleet vs. Video Distribution

- **Server Fleet (Fixed Monthly):** Hetzner dedicated servers hosting LiveKit SFU, TigerBeetle, Redis, Soketi, and egress workers are fixed-cost (~€90–€365/month). Server hardware costs are predictable and modest.
- **Video Distribution (Variable Metered COGS):** Spectator video bandwidth is metered:
  - Cloudflare Stream runs ~$1.00 per 1,000 delivered viewer minutes.
  - Enterprise CDN bandwidth runs ~$0.005–$0.015 per GB under negotiated commitment.
  - Self-hosted Mode B nodes (Frankfurt & Singapore) operate on fixed 1 Gbps / 10 Gbps unmetered bandwidth pipes, providing a predictable cost ceiling as volume scales. UPDATED **Self-hosted regional edges provide more predictable server-side cost, but capacity is bounded by uplink speed, peering quality, cache hit rate, regional load, operational monitoring, and failover behavior. This remains NOT_PROVEN until load tested.**

12.2 Monthly Cost Projections Across Deployment Tiers

Deployment Tier	Concurrent Audience	Server Fleet Cost (Fixed Monthly)	Video Delivery (Cloudflare Stream: 30 1-hr shows/mo)	Video Delivery (Enterprise CDN Egress: 30 1-hr shows/mo)	Total Estimated Cost (Stream Model)	Total Estimated Cost (Enterprise CDN / Mode B)
Launch	500 – 2,000	~€92 / mo	$900 – $3,600 / mo	$135 – $540 / mo	~$1,000 – $3,700 / mo	~$230 – $640 / mo
Growth	2,000 – 10,000	~€115 / mo	$3,600 – $18,000 / mo	$540 – $2,700 / mo	~$3,700 – $18,100 / mo	~$655 – $2,820 / mo
Regional	10,000 – 50,000	~€231 / mo	$18,000 – $90,000 / mo	$2,700 – $10,800 / mo	~$18,250 – $90,250 / mo	~$2,950 – $11,050 / mo
Mega Scale	50,000 – 150,000+	~€365 / mo	$90,000 – $270,000 / mo	$10,800 – $24,300 / mo	~$90,400 – $270,400 / mo	~$11,200 – $24,700 / mo

13. Execution Roadmap: Remaining Proof Gates

Before deploying unproven subsystems to production, each component must pass a dedicated verification gate:

Proof Gate	Subsystem Under Test	Success Criteria & Verification Seam	Status
Gate 1	**CDN / HLS Packaging Proof**	LiveKit Egress compositor packaging 720p 30fps CMAF chunks; verified playlist playback with <2s latency.	`NOT_PROVEN`
Gate 2	**Delivery-Cost Vendor Decision**	Commercial quote analysis: Cloudflare Stream vs. Enterprise CDN vs. Mode B self-hosted edge nodes.	`NOT_PROVEN`
Gate 3	**TigerBeetle Transfer-Chain Proof**	Dockerized TigerBeetle cluster executing explicit two-leg linked transfers with zero overdraft under 10k simulated concurrent gifts.	`NOT_PROVEN`
Gate 4	**PAG / libpag Legal & Build Proof**	Legal audit of libpag third-party license bundle; memory benchmarking (<40MB) and leak validation on Android and iOS.	`NOT_PROVEN`
Gate 5	**HLS-to-WebRTC Hot-Swap Proof**	Flutter `video_player` to `RTCVideoRenderer` cross-fade transition without black frames or audio glitches.	`NOT_PROVEN`
Gate 6	**Moderation Pipeline Proof**	End-to-end audit logging of host mute/kick actions in PostgreSQL and snapshot buffer storage for review.	`NOT_PROVEN`
Gate 7	**Thermal Adaptation Proof**	Physical battery temp monitor triggering dynamic bitrate and overlay degradation on low-end device under load.	`NOT_PROVEN`
Gate 8	**Production Route Proof**	Authoritative REST/WebSocket endpoints for room tokens, guest promotion, and battle state transitions.	`NOT_PROVEN`
Gate 9	**Failover / Reconnect Proof**	Network disconnect/reconnect tests validating graceful resumption without stream termination.	`NOT_PROVEN`

14. Conclusion & Planning Contract Notice

> **Important: Planning Contract Notice:**
> This document constitutes an **architectural planning contract and implementation specification**. It identifies proven foundations (`VERIFIED_PHYSICAL` Vanguard 720p30 camera and `VERIFIED_SOURCE` LiveKit WebRTC peer subscriptions) and strictly bounds unproven subsystems as `NOT_PROVEN` or `PLANNED`.
>
> Production readiness requires each numbered gate in Section 13 to pass mechanical and physical verification in its own dedicated work package.

15. Addendum: TikTok-Competitive Feature Alignment & Store-Compliant Operations

This addendum defines the product-level engagement, wallet compliance, and moderation mechanics required for competitive parity with leading platforms, grounded strictly in ConnectsApp system invariants.

★ NEW SECTION ADDED FOR TEAM REVIEW SECTION 15 RATIFIED ADDITIONS

TikTok-Competitive Feature Alignment & Store-Compliant Operations

The 7 strategic and operational capabilities below (Points 15.1 through 15.7) were newly added to the master livestream architecture specification. All new requirements are highlighted in yellow cards for immediate visibility.

POINT 15.1 Live Swipe Feed & Stream Pre-Warming

P1 LAUNCH REQUIREMENT

- **Objective & Baseline:** Fast swipe startup when navigating vertically between active livestreams is a ConnectsApp engineering target (`NOT_VERIFIED` with respect to proprietary third-party internal timings or guarantees).
- **Pre-Warm Pipeline:**
  - Background pre-initialization of the Flutter video player surface for the next candidate room in the feed.
  - Manifest prefetching (LL-HLS playlist) and initial segment/chunk caching for adjacent active stream candidates.
  - **Device Governance & Safety Fallback:** Governed by `ThermalGuardService`. Under elevated thermal conditions, high memory pressure, or on low-tier mobile hardware, prefetching is automatically disabled, falling back to standard on-demand startup.

POINT 15.2 Cross-Room Leaderboards & Rankings

P2 / V1.5 MILESTONE

- **Scope:** Platform-wide hourly, daily, and regional creator rankings are scheduled for V1.5 and are not a launch blocker (`NOT_VERIFIED` with respect to external platform leaderboard names or reset schedules).
- **Ledger-Backed Ranking Integrity:**
  - Leaderboard points are **strictly derived from settled TigerBeetle transfer events**, never raw client taps or unverified WebSocket triggers.
  - Asynchronous workers consume settled transaction streams to update Redis sorted sets (`ZINCRBY`) or materialized database views for low-latency ranking presentation.
- **Fraud & Anti-Manipulation Controls:**
  - Rapid velocity anomaly detection flags suspicious ranking spikes.
  - Payouts and ranking promotions remain subject to administrative compliance review before diamond release.

POINT 15.3 Creator Live Goals

P1 LAUNCH REQUIREMENT

- **In-Stream Crowdfunding HUD:** Creators can mount an interactive target milestone (e.g., target gift count or diamond threshold for an agreed on-stream activity).
- **Two-State Separation:**
  - **Hot Visual State (Redis):** Incremented in real time upon verified gift events and broadcast to all room viewports via the Soketi livestream event bus.
  - **Financial & Audit Truth (TigerBeetle & SQL):** All underlying coin-to-diamond transfers remain exclusively in TigerBeetle. Completed milestones are persisted asynchronously to cold SQL storage for creator analytics and historical audit.

POINT 15.4 Store-Compliant Account Wallet Strategy

P0 LAUNCH BASELINE

- **Unified Account-Level Wallet:** Users maintain a single account-level Coin balance accessible across supported platforms where permitted.
- **Server-Driven Payment Policy Service:** Dynamically resolves allowable purchase routes based on client platform (iOS, Android, Web), storefront region, distribution channel, user entitlement status, and age/KYC tier:
  - **Web-First Preferred Route:** Direct web wallet top-up is favored where permitted by applicable law and store terms.
  - **Platform In-App Billing:** Apple In-App Purchase (IAP) and Google Play Billing are utilized where required by platform agreements or strategically targeted.
  - **Alternative Billing Programs:** Approved external purchase or alternative billing workflows are integrated strictly where officially sanctioned under regional regulatory frameworks (e.g., EU DMA).
  - **Safe Degradation:** If no compliant purchase route exists for a given platform, country, or user category, coin purchase UI is cleanly disabled.
  - **Strict Compliance Invariant:** Prohibits in-app anti-steering tricks, hidden QR-code redirects, undocumented webviews, or misleading app-review behavior.
- **Ledger Flow Decoupling:**
  - TigerBeetle credits user Coin accounts **only after** cryptographically validated receipt validation, gateway webhook confirmation, or platform server notification.
  - **Fiat Coin Purchase** and **In-Stream Gift Spending** are strictly isolated, idempotent double-entry ledger flows.
- **Creator Payout Lifecycle:** Accrued creator Diamonds are held in pending reserve, subject to a mandatory cooling-off window, refund/chargeback reserves, identity verification, and anti-fraud approval before cash payout eligibility.

POINT 15.5 Audio-Only Multi-Guest Seats

P0/P1 LAUNCH REQUIREMENT

- **Flexible Stage Inclusion:** Approved multi-guests may join the interactive stage with microphone only; camera activation is optional.
- **Media & Bandwidth Architecture:**
  - Delivered as an Opus audio track via LiveKit SFU with camera track publishing unstarted or disabled.
  - Drastically reduces mobile upstream/downstream bandwidth and ISP/GPU silicon strain compared to multi-stream video.
- **Interactive HUD & Moderation:**
  - Renders guest profile avatar with real-time Voice Activity Detection (VAD) audio rings and mute status indicators.
  - Host retains full sovereign authority: remote one-tap mute, seat dismissal (kick), room ban, and abuse reporting.

POINT 15.6 Post-Stream Analytics & Highlight Clip Export

P1 / P2 PHASED ROLLOUT

- **Post-Stream Summary Card (`P1` Launch):**
  - Immediate broadcast summary presented to the creator upon stream conclusion.
  - Metrics: Total Diamonds received, estimated net payout, Peak Concurrent Viewers (PCV), total unique viewers, new followers gained, average watch duration, top contributors, and battle results.
- **Automated Highlight Slicing (`P1`/`P2` Phased):**
  - Highlight candidate timestamps flagged by event telemetry: gifting velocity spikes, chat message bursts, battle conclusion moments, and creator manual markers.
  - Sliced video clips are stored and served via static CDN / object storage (Cloudflare R2 / Hetzner Storage), completely isolated from the live low-latency video delivery pipeline.

POINT 15.7 Safety, Moderation & Host Governance

P0 LAUNCH BASELINE

- **Mandatory Operational Foundation:** Essential for platform launch compliance, user safety, and brand protection:
  - **Role-Based Permissions:** Host, co-host, designated room moderators, and platform compliance administrators.
  - **In-Stream Enforcement:** Immediate one-tap mute, guest seat ejection, room ban, and standardized viewer reporting workflows.
  - **Automated Content Filtering:** Server-side chat blocklists, regex-based profanity masking, and rate-limiting against message flooding.
  - **Eligibility & Velocity Safeguards:** Age verification barriers (18+ for gifting and broadcasting), guest invite restrictions (followers-only or verified-only), and strict velocity caps to prevent credit card testing and money laundering.
  - **Audit Logging:** Append-only database audit logs for all moderation interventions, bans, and anomalous gifting surges.